

Name: \_\_\_\_\_

## In-Class Activity

PS200C: Causal Inference in the Social Sciences  
Spring 2026

### 1. What is the causal question?

Which of the following best captures the causal question in the potential outcomes framework?

- (a) Compare average outcomes between treated and untreated groups
- (b) Compare outcomes for the same unit under treatment vs. control
- (c) Compare treated units to similar control units
- (d) Compare before-and-after outcomes for treated units

### 2. Observability

For a given unit  $i$ , classify each as *always observable*, *sometimes observable*, or *never observable*:

- $Y_i(1)$  and  $Y_i(0)$  (both at once)
- $D_i$
- $Y_i$
- $\tau_i = Y_i(1) - Y_i(0)$

### 3. Difference in Means

Consider:

$$E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0]$$

Which is correct?

- (a) Always equals ATE
- (b) Always equals ATT
- (c) Equals ATT plus selection bias
- (d) Equals ATC

### 4. Direction of Bias

Suppose:

- $Y_i(1) = Y_i(0)$  for all  $i$  (no treatment effect)
- Treatment is more likely for units with higher  $Y_i(1)$

What do you expect for:

$$E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0] ?$$

- (a) Zero

- (b) Positive
- (c) Negative

5. **Science Table**

| $i$ | $D_i$ | $Y_i(1)$ | $Y_i(0)$ |
|-----|-------|----------|----------|
| 1   | 1     | 10       | 4        |
| 2   | 1     | 1        | 2        |
| 3   | 0     | 3        | 3        |
| 4   | 0     | 5        | 2        |

Compute:

- ATE:
- ATT:

6. **Bonus:**

Why does

$$E[Y_i(1) \mid D_i = 1] = E[Y_i \mid D_i = 1]$$

hold?